VisionGuard Smart Cane

Complete project guide: A-to-Z build procedure, concepts, code, design decisions and viva questions



BACSE324 Embedded Systems Architecture, Course Project.

How to use this guide

1. **Tonight:** read Part 1 (the story) and Part 2 (concepts). Then read the A–Z steps once.
2. **Next:** study Part 4 (code) and Part 5 (design decisions). These are what faculty dig into.
3. **Before the review:** go through Part 6 (viva questions) out loud and look at the cheat sheet on the last page.

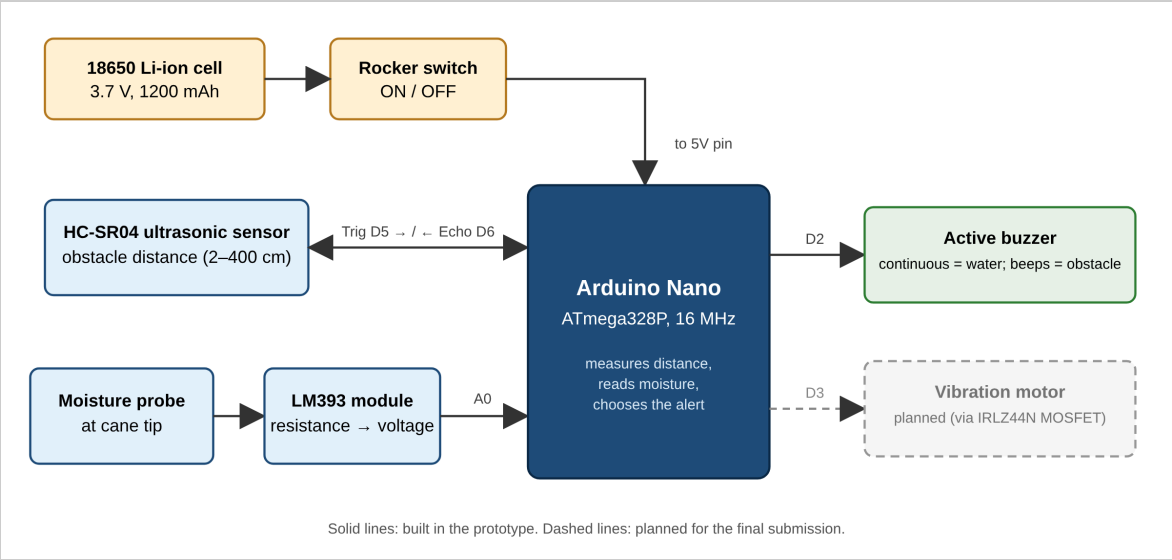
About the procedure: Part 3 describes the standard procedure for building this cane, matched to the actual hardware in the photos. Steps marked ◆ *standard practice* are the normal way such projects are tested (for example, testing on a breadboard first). Check with your teammate exactly which ones were done, so that everything you say is true. Faculty respect "this is how it works and why" far more than a story you can't back up.

Part 1: The project story in 2 minutes

"A white cane detects obstacles only when it touches them, and it can't warn about puddles or open drains. Our idea started as a **smart shoe** *with an ultrasonic sensor and a moisture sensor. We moved it to a* **cane**, *because the cane tip reaches water before the foot does, and a sensor on the shaft sees ahead more steadily than one on a moving shoe.*

The cane uses an **Arduino Nano** *as the brain. An* **HC-SR04 ultrasonic sensor** *measures distance by timing a sound echo. A* **moisture probe** *at the tip senses water, because water conducts electricity. A* **buzzer** *gives alerts: a* **continuous tone for water**, *which has top priority, and* **beeps that get faster** *as an obstacle comes closer than 50 cm. It runs on one* **18650 battery** *with an on/off switch, and the cane hardware costs about* **₹876**.

We tested each part, wrote and uploaded the code, mounted everything on a PVC pipe, connected the sensors with jumper wires on header pins and soldered the power wires. Next we plan to add a vibration motor, regulated power and more sensors."



Block diagram: inputs on the left, the Nano in the middle, outputs on the right

Part 2: Concepts you must understand

Concept	Simple explanation	Where it appears in our project
Voltage (V)	Electrical "pressure" that pushes current, like water pressure in a pipe	Battery 3.7 V; logic HIGH ≈ supply voltage
Current (A, mA)	How much electricity flows, like water flow rate	Whole cane draws about 40 mA
Ground (GND)	The 0 V reference point. All parts must share one GND, or signals have no common reference	All module GNDs join the Nano's GND
Microcontroller	A tiny computer on one chip (CPU + memory + input/output pins) that runs one program forever	ATmega328P on the Nano
Digital signal	Only two levels: HIGH (1) or LOW (0)	Trig, Echo and buzzer pins
Analog signal	Any voltage in a range, e.g. 0–5 V	Moisture module output on A0
ADC	Analog-to-Digital Converter: turns a voltage into a number. The Nano's is 10-bit: 2^{10} = 1024 levels, so 0–1023	`analogRead(A0)`
Pulse / pulse width	A short HIGH signal. Its width (how long it stays HIGH) can carry information	Echo pulse width = sound travel time
Ultrasound	Sound above human hearing (above 20 kHz)	HC-SR04 uses 40 kHz
Resistance (Ω)	How much a material opposes current. Dry air is very high; water with salts is much lower	Moisture probe
Voltage divider	Two resistances in series; the middle point's voltage depends on their ratio	Probe + fixed resistor in the LM393 module
Comparator (LM393)	Compares two voltages and outputs HIGH/LOW. The potentiometer sets the reference	D0 output (we don't use it; we read A0)
Active vs passive buzzer	Active: has a built-in oscillator and beeps on plain DC. Passive: needs a changing signal (`tone()`)	Ours is active: `digitalWrite(HIGH)` = sound
Capacitor	Stores charge like a tiny tank and smooths sudden voltage dips. Electrolytic ones have polarity (stripe = −)	220 µF, 50 V across 5V and GND
MOSFET	An electronic switch. A small pin signal controls a bigger current	IRLZ44N, for the planned vibration motor
Baud rate	Speed of serial communication in bits per second	Serial Monitor at 9600
Sketch	An Arduino program. `setup()` runs once at power-on; `loop()` repeats forever	visionguard.ino
Li-ion cell	Rechargeable lithium battery: 4.2 V full, 3.7 V nominal, about 3.0 V empty	18650, 1200 mAh, 4.44 Wh

Part 3: The A-to-Z build procedure

Phase 1: Idea and design (A–J)

A — **Identify the problem**

A white cane detects obstacles only on contact (about 1 m away), gives no distance information and cannot warn about water, puddles or open drains. WHO estimates at least 2.2 billion people have a vision impairment.

B — **Study existing solutions**

White cane (cheap but contact-only), guide dog (expensive, scarce), commercial smart canes (costly, imported, hard to repair). The gap: a low-cost aid that warns about obstacles *and* water.

C — **Define the objectives**

Detect obstacles within 50 cm; show distance through the beep rate; detect water at the tip with top priority; be portable and low-cost (one battery, under ₹900).

D — **First concept: smart shoe**

The first design was a smart shoe with an Arduino, an HC-SR04, a soil-moisture sensor, a buzzer (D2) and a vibration motor (D3). Its circuit used the same pins as the final cane.

E — **Redesign as a cane**

Moved to a cane because: (1) the tip touches water *before* the foot does; (2) a sensor on the shaft is higher and steadier than one on a stepping shoe; (3) there is room for the battery and electronics, away from the ground and sweat; (4) blind users already know how to use a cane.

F — **Draw the block diagram**

Inputs (ultrasonic sensor, moisture probe) → processor (Arduino Nano) → output (buzzer), plus the power block (battery → switch → 5V).

G — **Choose the controller: Arduino Nano**

Same ATmega328P chip as the Uno in the shoe design, but only 18 × 45 mm, so it fits on a cane. It has 5 V logic, a 10-bit ADC and USB programming, and costs about ₹250.

H

Choose the sensors

HC-SR04 ultrasonic sensor (cheap, 2–400 cm, works in darkness, unaffected by colour). Resistive soil-moisture probe with an LM393 module (cheap, gives an analog wetness value).

I

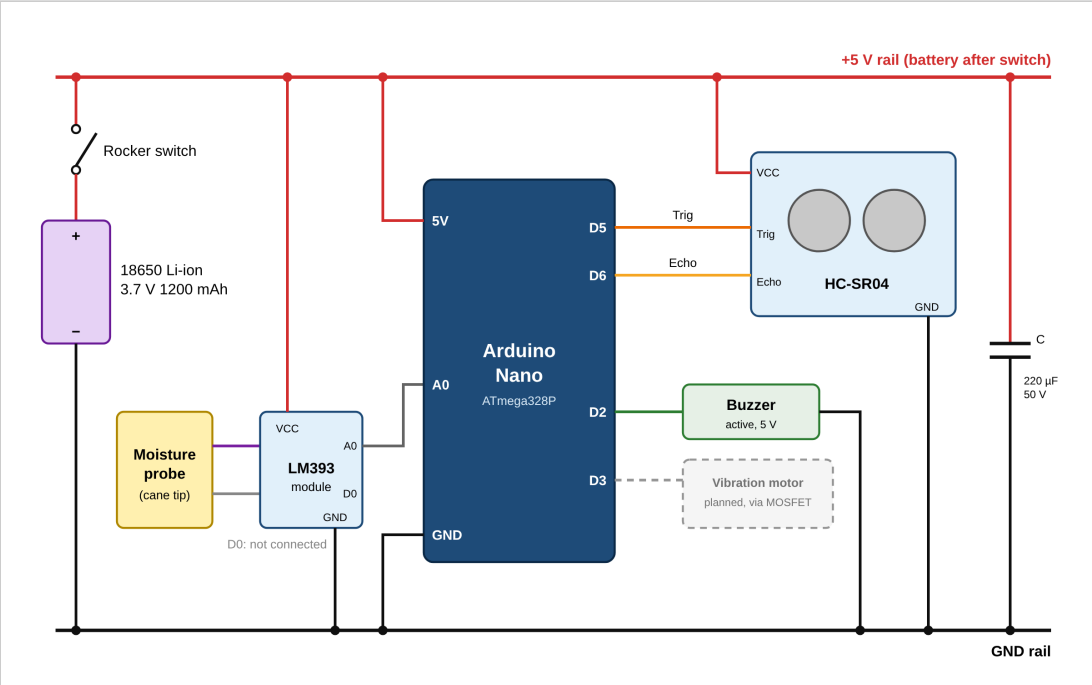
Choose the output and power parts

Active 5 V buzzer (no tone code needed). 18650 Li-ion cell (rechargeable, high capacity, common). Rocker switch (on/off). Electrolytic capacitor (supply smoothing). The IRLZ44N MOSFET and fuse were bought for later upgrades.

J

Make the pin plan and circuit diagram

Trig → D5, Echo → D6, moisture A0 → A0, buzzer → D2, D3 reserved for vibration. All VCCs on the 5V rail, all GNDs together.



Final circuit diagram

Phase 2: Purchase and testing (K–R)

K **Buy the components**

From Vellore Robo Electronics (invoice 3293, 07-10-2026), total ₹1,735 including 18% GST. This included the Nano, HC-SR04, moisture sensor, 2 × 18650 cells, holder, rocker switch, capacitors, jumpers, berg strips, wire, breadboard, soldering iron, solder, flux, MOSFET, fuse and a JST connector.

L **Install the software**

Arduino IDE 2.x. Clone Nanos use the CH340 USB chip, so install the CH340 driver if no COM port appears. Settings: Board = Arduino Nano, Processor = ATmega328P (or "Old Bootloader" if upload fails), Port = the new COM port.

M **Test the Nano** ◆ *standard practice*

Upload the built-in *Blink* example (File → Examples → Basics → Blink). The on-board LED (pin 13) flashing proves the board, cable and upload work.

N **Test the HC-SR04 on the breadboard** ◆ *standard practice*

Connect VCC → 5V, GND → GND, Trig → D5, Echo → D6. Upload the code and open the Serial Monitor at 9600. Move a hand closer and check that the distance falls.

O **Test the moisture sensor** ◆ *standard practice*

Probe → LM393 module; module VCC → 5V, GND → GND, A0 → A0. Dry should read about 1000+. Dip in water: the value drops a lot.

P **Test the buzzer** ◆ *standard practice*

Buzzer + (longer leg) → D2, − → GND. It should beep when a hand is closer than 50 cm. If it's silent, check the polarity and that it's an active buzzer.

Q **Write the full program**

Combine everything: measure distance, read moisture, apply priority logic (water → continuous; obstacle → graded beeps; else off), print values to Serial. See Part 4.

R **Calibrate the thresholds**

Note the dry and wet readings on the Serial Monitor and set `WET_SOIL_THRESHOLD` between them (default 500). The obstacle distance is set to 50 cm.

Phase 3: Assembly (S–W)

Layout from bottom to top: **probe → LM393 → HC-SR04 → Nano + buzzer → switch → battery**. The heavy parts sit near the handle so the cane is easy to swing.

S

Prepare the cane body

Cut a PVC pipe to cane length (light, cheap, strong, hollow). Clean it and mark where each part goes.

T

Mount the moisture probe and LM393 module

Hot-glue the probe at the bottom tip, with the copper prongs exposed and pointing down. Run its 2 wires (purple and grey) up to the LM393 module glued on the pipe. Module VCC (red) → 5V, GND (brown) → GND, A0 (grey) → Nano A0. D0 is left empty.



U

Mount the ultrasonic sensor

Fix the HC-SR04 on a foam block, facing forward in the walking direction, with nothing blocking its "eyes". Connect 4 jumpers: VCC, Trig (D5), Echo (D6), GND.

V

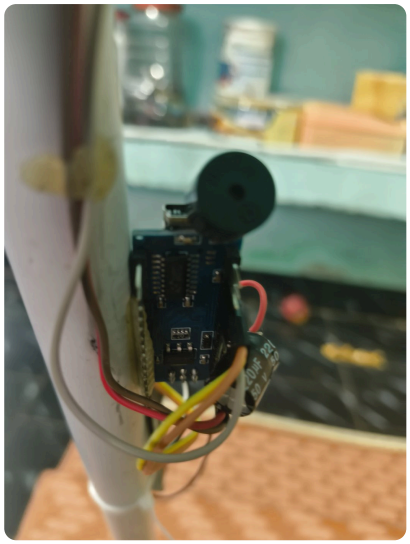
Mount the Nano and connect the wires

Hot-glue the Nano with its USB port reachable for reprogramming. Solder male header pins onto it, then plug the sensors' female jumper wires onto the pins: Trig → D5, Echo → D6, moisture A0 → A0. Solder the power wires (5V, GND), the buzzer (+ → D2, − → GND) and the 220 µF capacitor directly onto the Nano. **Why solder rather than use a breadboard?** A cane is swung and tapped constantly, so breadboard wires would shake loose. Soldered joints are permanent and compact.

W

Build the power circuit

Cable-tie the battery holder near the handle and glue the rocker switch within thumb reach. Battery + (red) → switch → Nano 5V. Battery − → GND. Capacitor across 5V and GND (stripe to GND). **The 5V pin, not VIN:** VIN goes through a regulator that needs about 7 V, and one cell gives only 3.7 V.



Phase 4: Finishing (X–Z)

X

Final upload and system test

With the switch OFF, connect USB and upload. Unplug USB, switch ON and run the tests: clear path = silent; notebook moving from 60 to 5 cm = beeps get faster; tip in water = continuous tone; water + obstacle together = continuous tone (priority).

Y

Safety and tidying

Insulate bare joints, tie the wires along the pipe, check that the battery polarity is correct, and make sure no bare battery wires can touch. Rule: **never connect USB with the switch ON** (USB 5 V would flow back into the cell).

Z

Document and present

Review 1 template, build guide, project report (with DFG and state diagram), presentation and code in the GitHub repository.

Part 4: The code, line by line

```
#define TRIG_PIN 5          // HC-SR04 Trig  → D5
#define ECHO_PIN 6          // HC-SR04 Echo  → D6
#define SOIL_PIN A0         // moisture module A0 → A0
#define BUZZER_PIN 2        // active buzzer → D2
#define VIB_PIN 3           // reserved for the vibration motor

const int OBSTACLE_DISTANCE_THRESHOLD = 50;   // cm
const int MIN_DISTANCE = 2;                   // HC-SR04 minimum
const int WET_SOIL_THRESHOLD = 500;           // below = water
const unsigned long ECHO_TIMEOUT_US = 30000;  // 30 ms
```

#define gives each pin a name, so the code is readable and the wiring can change in one place. **const** values are settings that never change while running.

```
void setup() {
  Serial.begin(9600);            // start the Serial Monitor link
  pinMode(TRIG_PIN, OUTPUT);     // Nano sends the trigger
  pinMode(ECHO_PIN, INPUT);      // Nano listens to the echo
  pinMode(BUZZER_PIN, OUTPUT);
  pinMode(VIB_PIN, OUTPUT);
  digitalWrite(BUZZER_PIN, LOW); // start silent
  digitalWrite(VIB_PIN, LOW);
}
```

setup() runs **once** at power-on. **pinMode** says whether each pin is an input or an output.

```
int readDistanceCm() {
  digitalWrite(TRIG_PIN, LOW);  delayMicroseconds(2);   // clean start
  digitalWrite(TRIG_PIN, HIGH); delayMicroseconds(10);  // 10 µs trigger
  digitalWrite(TRIG_PIN, LOW);
  unsigned long duration = pulseIn(ECHO_PIN, HIGH, ECHO_TIMEOUT_US);
  return duration * 0.034 / 2;   // cm (0 if no echo)
}
```

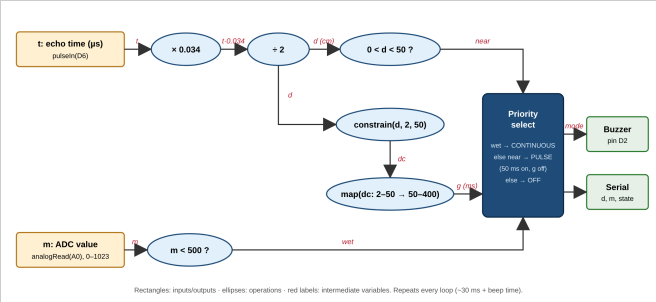
The 10 µs pulse tells the sensor to fire. **pulseIn** measures how many microseconds Echo stays HIGH (the round-trip time). Distance = time × 0.034 cm/µs ÷ 2. The **30 ms timeout** means that with no echo, it returns 0 instead of waiting the default 1 second.

```
void loop() {
  int distance = readDistanceCm();
  int soilMoisture = analogRead(SOIL_PIN);       // 0–1023
  // ... print distance, moisture and state to Serial ...

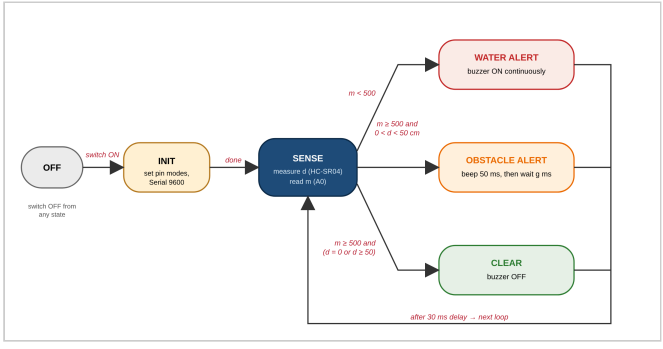
  if (soilMoisture < WET_SOIL_THRESHOLD) {        // PRIORITY 1: water
    digitalWrite(BUZZER_PIN, HIGH);              // continuous tone
    digitalWrite(VIB_PIN, HIGH);
  }
  else if (distance > 0 && distance < OBSTACLE_DISTANCE_THRESHOLD) { // PRIORITY 2
    int clamped = constrain(distance, MIN_DISTANCE, OBSTACLE_DISTANCE_THRESHOLD);
    int delayTime = map(clamped, MIN_DISTANCE, OBSTACLE_DISTANCE_THRESHOLD, 50, 400);
    digitalWrite(BUZZER_PIN, HIGH); digitalWrite(VIB_PIN, HIGH);
    delay(50);                                   // 50 ms beep
    alertsOff();
    delay(delayTime);                            // gap: 50–400 ms
  }
  else {                                         // path clear
    alertsOff();
  }
  delay(30);
}
```

Line	Meaning
`analogRead(A0)`	ADC converts the probe voltage into 0–1023. Dry ≈ 1023, wet = lower
`if (soil < 500)`	Checked **first**, so water always wins
`distance > 0`	0 means no echo, so it's ignored and doesn't count as an obstacle
`constrain(d, 2, 50)`	Keeps d inside 2–50, so map() never gives a strange value
`map(d, 2, 50, 50, 400)`	Linear scaling: 2 cm → 50 ms gap (fast beeps), 50 cm → 400 ms gap (slow beeps)
`delay(30)`	Short pause before the next reading so stray echoes die out

The full program is in `firmware/visionguard/visionguard.ino`. The original version is kept unchanged in `firmware/original/`.



Data flow graph



State diagram

Part 5: Design decisions and why

Decision	Alternatives	Why we chose it
Cane instead of shoe	Smart shoe (first idea)	Tip reaches water first; steadier, higher sensor; space for electronics; users already know canes
Arduino Nano	Uno, ESP32, Raspberry Pi	Small enough for a cane, cheap, 5 V logic, built-in ADC. A Pi is overkill (cost, power, boot time); an ESP32 uses 3.3 V logic and Wi-Fi isn't needed
Ultrasonic sensor	IR sensor, LiDAR	IR is affected by sunlight and object colour and has a short range; LiDAR is expensive. Ultrasonic is cheap, 2–400 cm and works in the dark
Resistive moisture probe	Capacitive sensor, water-level sensor	Cheapest and gives an analog value. Capacitive is better long-term (no corrosion), so it's in future scope
Reading A0, not D0	D0 with potentiometer	A0 gives the actual level, so the threshold is tuned in code and visible on Serial
Active buzzer	Passive buzzer, speaker	Beeps on a simple HIGH; no tone code or extra parts
Water before obstacle	Obstacle first	The tip is already at the danger spot, and slipping or falling into a drain is the bigger risk
50 cm alert distance	100–150 cm	Simple to demonstrate. It's a constant in code and can be raised after field testing (future scope)
Faster beeps when closer	Single beep, voice	Gives distance information without words, and is easy to learn
18650 on the 5V pin	9 V battery into VIN, 2 cells into VIN	One cell is light and rechargeable. VIN needs about 7 V, so one cell must use 5V. A boost converter is the planned upgrade
Soldering	Breadboard	A cane vibrates constantly; breadboard wires would come loose
PVC pipe	Aluminium cane	Cheap, light, easy to glue onto, and an insulator (no short circuits)

Part 6: Viva questions and answers

A. Basic questions

1. What is your project?
A smart cane for visually impaired people. It warns about obstacles within 50 cm (faster beeps when closer) and gives a continuous tone when the tip touches water.

2. What problem does it solve?
A white cane detects obstacles only on contact, gives no distance information and can't detect water or drains.

3. Who is it for?
Blind and low-vision people, and elderly people with poor eyesight.

4. What are the main components?
Arduino Nano, HC-SR04 ultrasonic sensor, soil-moisture probe with LM393 module, active buzzer, 18650 battery, rocker switch and a capacitor, mounted on a PVC pipe.

5. What is the novelty?
It combines obstacle detection *and* water detection at very low cost, with a priority system.

6. How much did it cost?
About ₹876 for the cane hardware; ₹1,805 in total including tools and spare parts.

7. Which programming language did you use?
Arduino C/C++, written and uploaded with the Arduino IDE.

8. What does the user hear?
Silence when clear; short beeps under 50 cm that speed up as the obstacle gets closer; a continuous tone for water.

9. How is it switched on?
With a rocker switch between the battery and the Nano's 5V pin.

10. What was your original idea?
A smart shoe. We moved it to a cane because the tip reaches water first and the sensor is steadier.

B. Components

11. Which microcontroller is on the Nano?
ATmega328P, an 8-bit AVR running at 16 MHz, with 32 KB flash, 2 KB SRAM and 1 KB EEPROM.

12. How many pins does the Nano have?
14 digital I/O pins (6 with PWM) and 8 analog inputs (A0–A7).

13. Why not an Arduino Uno?
Same chip, but the Nano is much smaller (18 × 45 mm) and lighter, so it suits a cane.

14. Why not a Raspberry Pi?
More expensive, uses more power, takes time to boot, and isn't needed for simple sensing.

15. What are the HC-SR04's pins?

VCC, Trig, Echo and GND.

16. What is the HC-SR04's range?

2 cm to 400 cm, with a beam of about 15° and an accuracy of about 3 mm in ideal conditions.

17. What frequency does it use?

40 kHz ultrasound, which is above human hearing (20 kHz).

18. What is the blue board with the potentiometer?

An LM393 comparator module. It converts the probe's resistance to a voltage (A0), and the pot sets the threshold for the digital output (D0).

19. Why didn't you use D0?

A0 gives the exact moisture level, so we set the threshold in software and can see the values on Serial.

20. Active or passive buzzer, and why?

Active. It has its own oscillator and beeps with a simple HIGH signal.

21. What is the capacitor for?

It smooths the supply so that buzzer switching doesn't cause voltage dips that could reset the Nano.

22. Battery specifications?

18650 Li-ion, 3.7 V nominal (4.2 V full, about 3.0 V empty), 1200 mAh, 4.44 Wh.

23. Why are the MOSFET and fuse on the bill?

For upgrades: the IRLZ44N MOSFET to drive a vibration motor, and the 1 A fuse to protect the battery wiring.

24. What is a breadboard for, and why isn't it on the cane?

It's for testing without soldering. The final cane uses female jumper wires on header pins and soldered power wires, because vibration would loosen breadboard wires.

25. Why PVC pipe?

It's light, cheap, strong, an electrical insulator and easy to glue onto.

C. Working principle and physics

26. How does the ultrasonic sensor work?

A 10 µs trigger makes it send eight 40 kHz pulses. Echo stays HIGH until the reflection returns. Distance = time × speed of sound ÷ 2.

27. Why divide by 2?

The sound travels to the object and back, so the measured time is for twice the distance.

28. Where does 0.034 come from?

The speed of sound is about 340 m/s = 34,000 cm/s = 0.034 cm/µs.

29. Work out the distance for an echo of 2000 µs.

2000 × 0.034 / 2 = 34 cm.

30. Does temperature affect it?

Yes. The speed of sound ≈ 331 + 0.6·T m/s, an error of about 0.2% per °C. It's small for our purpose.

31. How does the moisture probe work?

Water with dissolved salts conducts electricity. When water bridges the two tracks, the resistance falls, so the voltage on A0 falls and the reading drops.

32. Why is dry about 1023?

Dry means very high resistance, so A0 is pulled up to nearly the supply voltage, the maximum of the 10-bit ADC.

33. What is ADC resolution?

10-bit means 1024 levels. At 5 V that's about 4.9 mV per step.

34. Does a weak battery change the moisture reading?

Very little. The divider and the ADC reference use the same supply, so the reading depends on the resistance ratio (it's ratiometric).

35. What surfaces does ultrasonic miss?

Soft surfaces (cloth, foam) absorb the sound; angled surfaces deflect it; thin poles may be too small to reflect enough.

D. Code

36. Difference between setup() and loop()?

setup() runs once at power-on; loop() repeats forever.

37. What does pinMode do?

It sets a pin as INPUT or OUTPUT.

38. What does pulseIn() return?

How long a pin stays HIGH (or LOW), in microseconds. We use it for the echo time.

39. Why the 30 ms timeout?

Without it, pulseIn waits up to 1 second when there's no echo, which would freeze the cane. 30 ms covers about 5 m round trip.

40. What does map() do?

It scales a value linearly from one range to another: 2–50 cm becomes a 50–400 ms gap.

41. Why constrain() before map()?

It keeps the input in range so map() doesn't produce values outside 50–400.

42. Why check distance > 0?

0 means no echo (timeout), which should count as "clear", not as an obstacle.

43. How is water given priority in code?

It's the first `if`, so the obstacle check runs only when it's not wet.

44. What is Serial.begin(9600)?

It starts serial communication at 9600 baud, so readings can be seen on the laptop.

45. What does #define do?

It gives a name to a value, replaced before compiling. It makes the code readable and easy to change.

46. What's a drawback of using delay()?

It blocks: the Nano can't read sensors during the delay (up to 450 ms). A millis()-based timer would fix this.

47. How fast does the loop run?

When clear, about 30–60 ms per loop (many readings per second). During obstacle beeps, up to about 0.5 s per loop.

48. How do you upload code?

USB cable, Arduino IDE, Board = Nano, Processor = ATmega328P, choose the port, click Upload. Keep the switch OFF while doing it.

E. Power and electrical

49. Why the 5V pin and not VIN?

VIN feeds the on-board regulator, which needs about 7 V or more. One 3.7 V cell is too low, so it goes to 5V directly.

50. Can the Nano run on 3.7 V?

Yes in our tests, but it's below the designed 5 V, so it gets less reliable as the battery drains. A boost converter is the proper fix.

51. How long does the battery last?

Theoretically 1200 mAh ÷ about 40 mA ≈ 30 hours. Real life will be less; we haven't measured it.

52. Why must the switch be OFF when using USB?

The battery is wired straight to the 5V rail, so USB 5 V would push current into the Li-ion cell.

53. How much current can a Nano pin give?

About 20 mA recommended, 40 mA absolute maximum. That's enough for the buzzer, but not for a motor.

54. How would you connect a vibration motor?

Through the IRLZ44N logic-level MOSFET as a switch (gate to D3), with a flyback diode across the motor to absorb voltage spikes.

55. Why must all grounds be common?

Signals are measured relative to GND. Without a shared ground, the Nano can't read the sensor signals correctly.

56. How do you charge the battery?

Remove it and use a proper Li-ion charger. A TP4056 module is a planned upgrade.

F. Challenging questions

57. Is 50 cm enough warning?

At walking speed (about 1 m/s) it gives about 0.5 s, which is short. It's one constant in the code; after field testing we'd raise it to 100–150 cm.

58. Can it detect stairs going down or potholes?

Not reliably. The sensor faces forward. A downward-angled second sensor is future scope.

59. Head-level obstacles like branches?

No. The sensor is on the shaft. An upward-angled sensor is planned.

60. What about false water alarms in the rain?

Rain on the probe can trigger it. That's acceptable, since wet ground is slippery anyway. Calibration and averaging reduce it.

61. Won't the probe corrode?

Yes, DC through water corrodes it. Fixes: power the probe only while reading, or switch to a capacitive sensor.

62. Can a blind user hear the buzzer in traffic?

Possibly not, which is why a vibration motor is the first planned upgrade.

63. How accurate is it?

The HC-SR04 is about ±3 mm in ideal lab conditions. On a moving cane, soft or angled targets reduce accuracy. We haven't done a formal accuracy test yet.

64. Two people using the cane near each other: interference?

Possible, since both use 40 kHz, and a stray echo could give a wrong reading. Averaging readings helps.

65. The datasheet suggests over 60 ms between pings. Does your code follow that?

Our loop waits 30 ms plus the measurement time, which can be under 60 ms when clear. Increasing the delay to 60 ms is a simple improvement.

66. Is it waterproof?

No. The electronics are exposed. A sealed enclosure is future scope.

67. What was the hardest part?

Answer honestly from your own experience. Common ones: choosing how to power it from one cell, finding the moisture threshold, and mounting the sensors so they don't wobble.

68. What did you learn?

Sensor interfacing (digital timing and analog ADC), priority logic, power design and the trade-offs of a real device.

69. How is this "embedded"?

A dedicated microcontroller runs one program continuously, reads sensors and controls outputs in real time, with no operating system.

70. If you had more time and money, what would you add?

Vibration motor, 5 V boost converter + charger + fuse, up/down sensors, capacitive probe, GPS + SOS message, and a PCB in a waterproof case.

> **If you don't know an answer:** don't guess. Say *"I'm not sure about that detail, but I'd check the datasheet / test it like this..."* Faculty value honesty and a sensible method more than a made-up answer.

Part 7: Demo plan

1. Show the parts along the cane, from the tip upwards: probe, LM393, HC-SR04, Nano + buzzer, switch, battery.
2. Switch ON: silence means the path is clear.
3. Bring a notebook slowly from 60 cm to 5 cm: the beeps get faster.
4. Dip the tip in a cup of water: continuous tone. Then hold the notebook close as well: still continuous (water priority).
5. Optional: switch OFF, connect the laptop and show live readings on the Serial Monitor.

> **Carry:** the cane, a charged battery, USB mini-B cable, laptop with the Arduino IDE and code, a cup of water and a tissue, and printed documents. **Backup:** if the battery is flat, switch OFF and power the cane from the laptop's USB.

Part 8: Glossary

ADC	Analog-to-Digital Converter	**GND**	Ground, the 0 V reference
VCC	Supply voltage pin	**I/O**	Input/Output
PWM	Pulse Width Modulation	**IDE**	Integrated Development Environment
µs	Microsecond (one millionth of a second)	**mAh**	Milliamp-hours (battery capacity)
DFG	Data Flow Graph	**MOSFET**	Electronic switch transistor
Baud	Serial speed in bits per second	**Flyback diode**	Protects against motor voltage spikes
CH340	USB-to-serial chip on clone Nanos	**Ratiometric**	Reading depends on a ratio, not on absolute voltage

Cheat sheet (read this before you walk in)

Key numbers		Pins	
Alert distance	50 cm	Trig	D5
Water threshold	< 500 of 1023	Echo	D6
Beep	50 ms on, 50–400 ms gap	Moisture	A0
Speed of sound	0.034 cm/µs (340 m/s)	Buzzer	D2
HC-SR04	40 kHz, 2–400 cm, ~15°	Vibration (planned)	D3
Trigger pulse	10 µs	Battery + (via switch)	5V
pulseIn timeout	30 ms	All grounds	GND
ADC	10-bit, 0–1023		
Battery	18650, 3.7 V, 1200 mAh		
Nano	ATmega328P, 16 MHz, 32 KB		
Cost	₹876 cane / ₹1,805 total		

Formula

d = t × 0.034 / 2

e.g. 1000 µs → 17 cm

Priority

1. Water → continuous
2. Obstacle < 50 cm → faster beeps
3. Else → silent

Why a cane, not a shoe

Tip reaches water first · steadier sensor · room for electronics · familiar to users

Future scope

Vibration motor · boost converter + charger · up/down sensors · capacitive probe · GPS SOS · PCB + enclosure

Never claim: vibration, GPS or Bluetooth are working (they're future scope). **Safety rule:** switch OFF before connecting USB.